

Title of the Proposal: Engineering Trustworthy and Intelligent Autonomous Systems: Explainability, Safety, Security and Responsible Innovation

A Brief Description of the Proposed Area

Artificial Intelligence, machine learning, robotics, computer vision, connected sensors and the Internet of Things are rapidly transforming healthcare, transportation, manufacturing, agriculture, communication, surveillance and critical infrastructure. The increasing autonomy of these systems has created a need for engineering solutions that are not only intelligent and efficient but also explainable, robust, secure, reliable and safe under real-world operating conditions. Challenges such as adversarial attacks, sensor failures, uncertain environments, biased training data, communication disruptions and unpredictable system behaviour must be addressed through advanced system design, testing and validation.

The proposed Special Session, “**Engineering Trustworthy and Intelligent Autonomous Systems: Explainability, Safety, Security and Responsible Innovation,**” will provide a technical platform for presenting emerging research on the design, development, integration and evaluation of dependable AI-enabled autonomous systems. The session will encourage research on explainable and interpretable machine learning, robust deep-learning architectures, intelligent control, sensor fusion, edge AI, fault detection, cybersecurity, privacy-preserving computation, human–machine collaboration and safety-aware system design

A central objective of the session is to examine how engineering principles can be incorporated throughout the lifecycle of intelligent systems—from data acquisition and model development to hardware implementation, deployment, monitoring and performance assessment. Particular attention will be given to methods for improving the reliability of AI models under changing environments, detecting abnormal system behaviour, protecting connected devices against cyberattacks and establishing effective human supervision over safety-critical automated operations. Responsible innovation, standards and regulatory requirements will be considered only insofar as they influence technical design, testing, certification and deployment.

The proposed session is closely aligned with the central theme of **Emerging Electronics and Automation**. It complements Track 1 by addressing intelligent control, robotics and autonomous-system safety; Track 2 through computer vision, multimedia authentication and deepfake detection; Track 3 through secure IoT architectures, wireless communication and edge intelligence; Track 4 through intelligent sensors, biomedical instrumentation and sensor fusion; Track 5 through hardware acceleration, embedded AI and energy-efficient intelligent devices; and Track 6 through trustworthy machine learning, explainable AI and robust deep-learning systems.

Unlike the existing tracks, which address individual technologies and their applications, this Special Session adopts a system-level engineering perspective. It brings together AI models, sensors, communication networks, control mechanisms, computing hardware and human supervision to develop integrated autonomous systems that remain dependable under uncertainty, faults and cyber threats. The session will therefore complement, rather than duplicate, the existing tracks by focusing specifically on the technical assurance, validation and trustworthy deployment of intelligent autonomous technologies.

Details of Session Chairs/Organizers

Organizer 1:

- **Name:** Dr. Amit Verma
- **Designation and Affiliation:** Associate Professor, School of Computer Science and Engineering and Advanced Centre of Research and Innovation (ACRI), CGC University, Mohali, Punjab, India
- **Photo:**



- **Brief Bio:** Prof. (Dr.) Amit Verma is an Associate Professor at the School of Computer Science and Engineering and the Advanced Centre of Research and Innovation, CGC University, Mohali. He has more than 15 years of academic and research experience and holds a PhD in Artificial Intelligence, with research focused on deep-learning-based plant disease detection. He has also undertaken postdoctoral research at IIT Roorkee. As a DST Young Scientist Fellow, he served as Principal Investigator for two Government of India-funded projects concerning AI applications in agriculture and AI awareness in rural and border districts, worth approximately ₹60 lakh. He has published more than 100 research articles, authored two books, and published four patents. A two-time GATE qualifier and NVIDIA-certified academic, he has also headed an AI Centre of Excellence and administered an NVIDIA DGX A100 facility.
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Organizer 2:

Name: Dr. Mukesh Kumar

Designation and Affiliation: Associate Professor, Department of Computer Applications, Chandigarh School of Business, CGC University, Mohali, Punjab, India

Photo:

**Brief Bio:**

Dr. Mukesh Kumar is an Associate Professor in the Department of Computer Applications, Chandigarh School of Business, CGC University, Mohali. He is also associated with the Advanced Centre of Research and Innovation as a postdoctoral researcher. He has more than 15 years of academic and research experience, including previous appointments at Lovely Professional University, Chitkara University and GGS College of Modern Technology. His principal research interests include machine learning, data mining, blockchain technology, ensemble learning, hybrid computational models, distributed systems, educational data mining and smart-city technologies. He has published numerous research papers in reputed journals and conference proceedings and has contributed as a book editor and patent researcher. Dr. Kumar actively mentors students, promotes interdisciplinary research collaborations and works on applying emerging technologies and data-driven methods to education and real-world problems.

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Organizer 3:

Name: Mr. Bineet Singh

Designation and Affiliation: Assistant Professor of Law, CGC University, Mohali, Punjab, India;
Doctoral Researcher, Himachal Pradesh University, Shimla

Photo:**Brief Bio:**

Mr. Bineet Singh is an Assistant Professor of Law at CGC University, Mohali, and a doctoral researcher at Himachal Pradesh University, Shimla. He has more than seven years of experience across legal research, teaching and industry. He holds an LL.M. with a specialization in Criminal Law from Lovely Professional University. His research interests include jurisprudence, legal philosophy, comparative constitutional law, public policy, constitutional governance and the legal regulation of emerging technologies. He previously worked as a Research Assistant at the High Court of Himachal Pradesh and has also contributed to teaching and research at Himachal Pradesh National Law University, Shimla. His research has been published in Scopus-indexed, former UGC-CARE-listed, peer-reviewed, and other reputed journals. He has also completed certified courses in common law, international criminal law, negotiation and conflict resolution.

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Technical Outline of the Session and Topics:

- Engineering architectures for trustworthy AI-enabled autonomous systems
- Explainable and interpretable artificial intelligence
- Robust machine learning and deep-learning architectures
- Intelligent control and adaptive control of autonomous systems

- AI-enabled robotics and industrial automation
- Autonomous vehicles, unmanned aerial vehicles and mobile robots
- Multi-sensor data fusion and intelligent perception
- Computer vision for autonomous navigation and monitoring
- Real-time object detection, tracking and scene understanding
- Edge AI and embedded intelligence for autonomous systems
- Hardware acceleration for machine-learning applications
- Energy-efficient AI and resource-constrained intelligent devices
- Secure IoT architectures and connected autonomous systems
- Cybersecurity of robotic, industrial and cyber-physical systems
- Adversarial machine learning and defence against AI attacks
- Privacy-preserving machine learning and secure computation
- Federated learning for distributed intelligent systems
- Fault detection, diagnosis and recovery in autonomous systems
- Resilient control under sensor, actuator and communication failures
- Reliability modelling and performance evaluation of AI systems
- Verification and validation of machine-learning models
- Safety-aware AI and risk-sensitive autonomous decision-making
- Human-in-the-loop and human-machine collaborative systems
- Digital twins for monitoring and validation of autonomous systems
- Predictive maintenance using AI and intelligent sensors
- AI-enabled biomedical instrumentation and healthcare systems
- Wearable, implantable and remote-monitoring technologies
- Deepfake detection and multimedia authentication technologies
- Detection and mitigation of algorithmic bias through technical methods
- Generative AI reliability, hallucination detection and model evaluation
- Large language models for engineering and automation applications
- AI-based anomaly detection in critical infrastructure
- Smart manufacturing and Industry 4.0/5.0 applications
- Intelligent transportation and smart-city technologies
- Sustainable AI and environmentally responsible system design
- Safety-by-design, security-by-design and privacy-by-design approaches
- Technical standards, testing protocols and certification of intelligent systems

- Compliance-aware engineering and responsible technological deployment

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Other Details:

The proposed session will follow the conference's prescribed paper-submission, plagiarism-screening and peer-review requirements. Every submission will be evaluated for originality, technical quality, methodological soundness, interdisciplinary relevance and alignment with the scope of the special session.

The organizers will promote the call for papers through academic institutions, professional networks, research groups and relevant IEEE communities. Depending upon the conference format and approval of the organizing committee, the session may also include an invited keynote or expert panel on the governance of AI and autonomous systems. At least one session chair from a technical discipline and one organizer with expertise in law or technology policy should be present to preserve the session's interdisciplinary character.